

using HTML5, CSS3 and JavaScript, and deploy it through the OpenMEAP interface.

Features

- OpenMEAP is available with an open source licence, which allows all organisations to take an active role in ensuring the security and quality of the product by providing inputs in the development of various features.
- It can be used to enable existing teams to create applications by using HTML 5, CSS3, and JavaScript for a device-agnostic solution or service offering. OpenMEAP mobile clients remove the need for developing mobile apps across various platforms such as iOS, Android, Windows and BlackBerry.
- It does not require any new ports to be opened in the firewall. This allows the product to be implemented without creating unnecessary vulnerabilities.
- OpenMEAP offers end-to-end security, as it supports 256 bits Advanced Encryption Standard (AES) and Secure Socket Layer (SSL). It offers full Public Key Infrastructure (PKI) agreement and also provides an option to share it with the user's existing Certificate Authority.
- Its architecture is designed for high availability; applications developed and designed using proper best practices can continue to function in a number of adverse situations, including network loss or power outages.
- The OpenMEAP admin interface manages application versions, device management, deployments and security policies. Administrators can also design a security profile that is appropriate for the implementation. It allows users to build, manage, test and install mobile applications.
- OpenMEAP supports simple object access protocol and Web services that allow access to other systems or services from the mobile application.

OpenMEAP installation

Download OpenMEAP installer from <http://www.openmeap.com/products/download/> based on the operating system you are using. The installer requires the Mac OS X 10.6+ 64 bits or Windows XP/Vista/7/8 or any Linux distribution.

Here is a list of OpenMEAP pre-requisites/software requirements:

- Eclipse IDE for Java EE developers (Windows or Mac)
- Windows Phone 8 SDK (Win 8)
- Android SDK (Windows or Mac)
- Xcode 4.2+ (Mac)
- Apache Tomcat 7.0
- Apache IvyDE
- Apache Ant 1.8+
- Java JDK 1.6+

One of the most promising advantages of using

OpenMEAP is an integrated and open source mobile technology solution with less administrative overhead; less need of upgrades that reduces management costs.

OpenMEAP provides flexibility to use any HTML5 or CSS3 framework or responsive design frameworks. It provides freedom to use any mobile user interface such as browsers supporting HTML5 and CSS3. It also provides freedom to deploy OpenMEAP in the cloud, hosted or in on-premise environment based on existing business strategy.

OpenMEAP avoids vendor lock-in and open source code will allow to modify features according to the business needs. Users can select integrated development environments of their own choice and they are not bound to use any specific IDE

Source code is available in public domain hence there is better chance to ensure security claims by rigorous auditing.

Figure 1: Benefits of OpenMEAP

OpenMEAP is the flexibility to deploy it in the cloud environment—either hosted or in an on-premise environment, based on existing business strategies.

OpenMEAP on the Amazon Web Services (AWS) public cloud

The OpenMEAP 1.4.5 version is available in the AWS Marketplace. It is available on Base Operating System Linux/UNIX and Amazon Linux 2012.03.3 as a 64-bit Amazon Machine Image (AMI). AMI is an encrypted machine image of a specific virtual machine. It contains a base OS, and a set of applications and services for achieving a specific purpose with a specific configuration. Amazon Web Services (AWS) is a public cloud service provider that provides a computing environment for running instances of an AMI on a pay-as-you-go basis. Amazon EC2 and Amazon EBS services are required to run open source OpenMEAP. There are multiple OpenMEAP bundles available in the AWS Marketplace such as OpenMEAP - Small Business Support Bundle [Red Hat and Amazon Linux]; OpenMEAP - Enterprise Support Bundle [Amazon Linux and Red Hat]; OpenMEAP - Developer Support Bundle [Amazon Linux and Red Hat]; and OpenMEAP - Mobile Enterprise Application Platform [Amazon Linux and Red Hat].

Here are the steps you need to follow to install OpenMEAP on AWS:

- Visit AWS Marketplace to access an OpenMEAP bundle
- Select an OpenMEAP version from AWS Marketplace to deploy it
- Configure it for an AWS server setup
- Configure firewall rules for http, https and ssh
- Start the OpenMEAP virtual machine on AWS
- Log in with admin account OpenMEAP
- Connect to EC2 instance with an ssh instance for any other required configuration or modifications

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